

Day 14: (Part 2): Subnetting

CLASS C NETWORKS

Subnets/Hosts (Class C)		
Prefix Length	Number of Subnets	Number of Hosts
/25	2	126
/26	4	62
/27	8	30
/28	16	14
/29	32	6
/30	64	2
/31	128	0 (2)
/32	256	0 (1)

Subnetting

1 1 0 0 0 . 0 . 0 . 0 . 0 . 1 0 1 0 1 0 0 0 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0
 192 . 168 . 255 . 0
 1 1 0 0 0 . 0 . 0 . 0 . 1 0 1 0 1 0 0 0 . 1 1 1 1 1 1 1 1 . 1 0 0 0 0 0 0 0
 192 . 168 . 255 . 128

Borrowing 1 bit = can make 2 subnets

2^x = number of subnets
 (x = number of 'borrowed' bits)

$2^n - 2$ = number of hosts
 (n = number of host bits)

CLASS B NETWORKS



Subnets/Hosts (Class B)

Prefix Length	Number of Subnets	Number of Hosts
/17	2	32766
/18	4	16382
/19	8	8190
/20	16	4094
/21	32	2044
/22	64	1022
/23	128	510
/24	256	254

Prefix Length	Number of Subnets	Number of Hosts
/25	512	126
/26	1024	62
/27	2048	30
/28	4096	14
/29	8192	6
/30	16384	2
/31	32768	0 (2)
/32	65536	0 (1)

